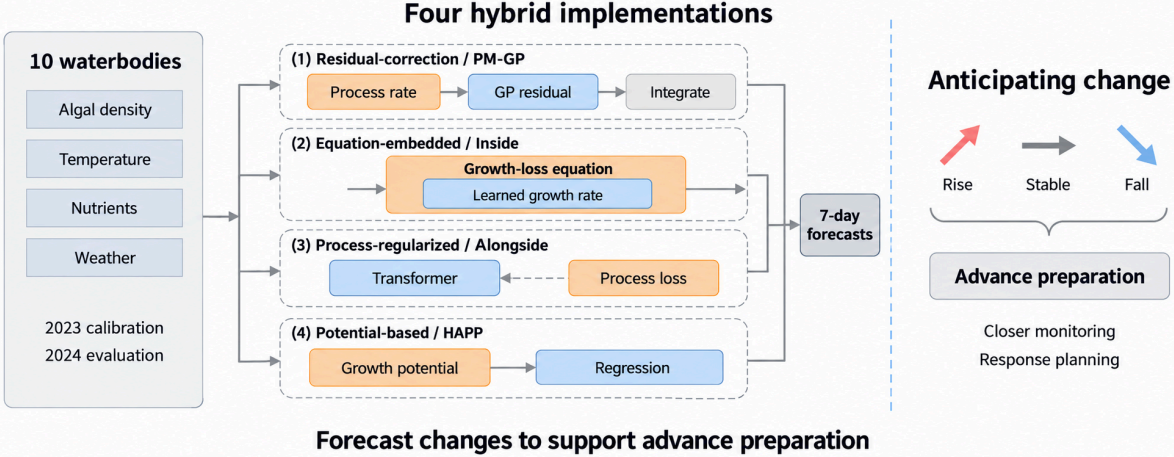

Four hybrid implementations
10 waterbodies
Algal density
Temperature
Nutrients
Weather
2023 calibration
2024 evaluation
(1) Residual-correction / PM-GP
Process rate
GP residual
Integrate
(2) Equation-embedded / Inside
Growth-loss equation
Learned growth rate
(3) Process-regularized / Alongside
Transformer
Process loss
(4) Potential-based / HAPP
Growth potential
Regression
7-day forecasts
Anticipating change
Rise
Stable
Fall
Advance preparation
Closer monitoring
Response planning
Forecast changes to support advance preparation